

CSCI 6355 Bioinformatics

Introduction

What is bioinformatics?

Bio – informatics = is conceptualizing biology in term of molecules and the applying “informatics” techniques (derived from disciplines such as math, cs, statistics) to understand and organized the information associated with these molecules, on a large-scale*

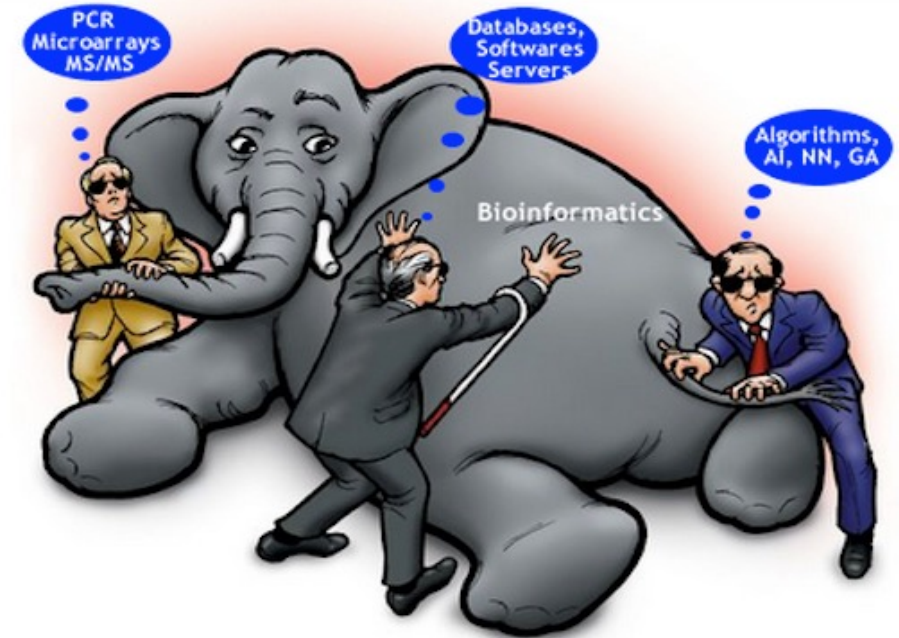
*Dr. M. Gerstein. Yale University

What is bioinformatics?

- Interdisciplinary research area that is the interface between the biological and computational sciences.
- Biological sciences – neurobiology, chemistry, molecular biology, medicine, biotechnology, biomedical engineering, etc.
- Goal – To uncover the wealth of biological information hidden in the mass of data and obtain a clearer insight into the fundamental biology of organisms.
- This knowledge have profound impacts on fields as varied as human health, agriculture, the environment, energy and biotechnology.

Why is bioinformatics important?

The greatest challenge facing the bioinformatics community today is the need for developing intelligent and efficient methods to store, retrieve, network, process, analyze, and visualize of this mass of data.



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Biological databases

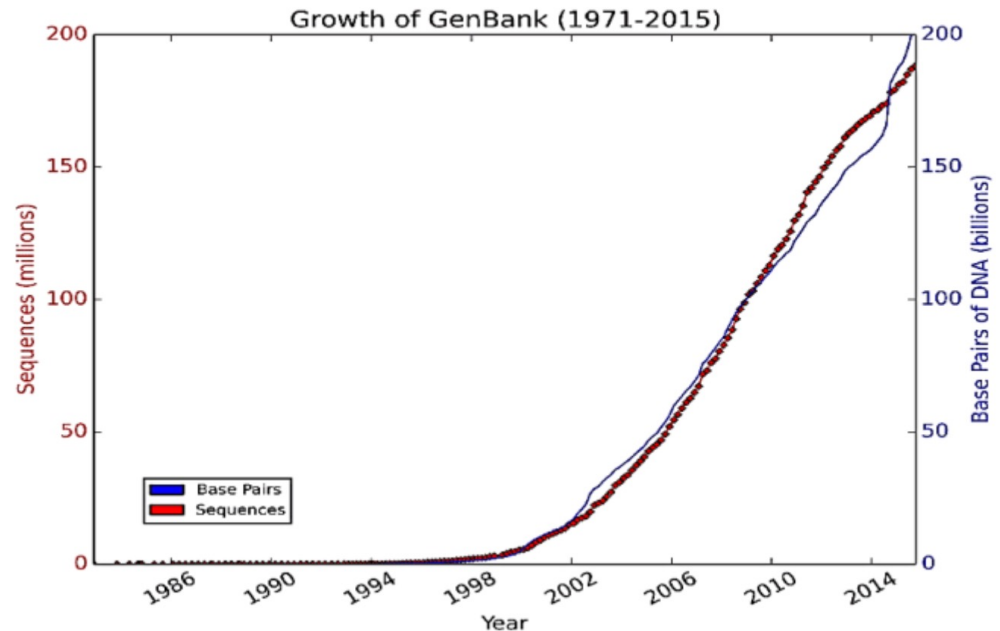
- Specialist databases for particular subjects have been set-up for example:
 - GenBank, DDBJ, ENA (INSDC)
 - for nucleotide sequence data
 - UniProt, Swiss-Prot
 - for protein sequence data
 - PDBe, PDBj
 - for protein structure databases
 - OMIM
 - Online catalog focusing on human genes and genetic disorders



Data growth in GenBank

A cumulative plot of the amount of DNA sequence (in bp) deposited in GenBank since 1971 to 2015.

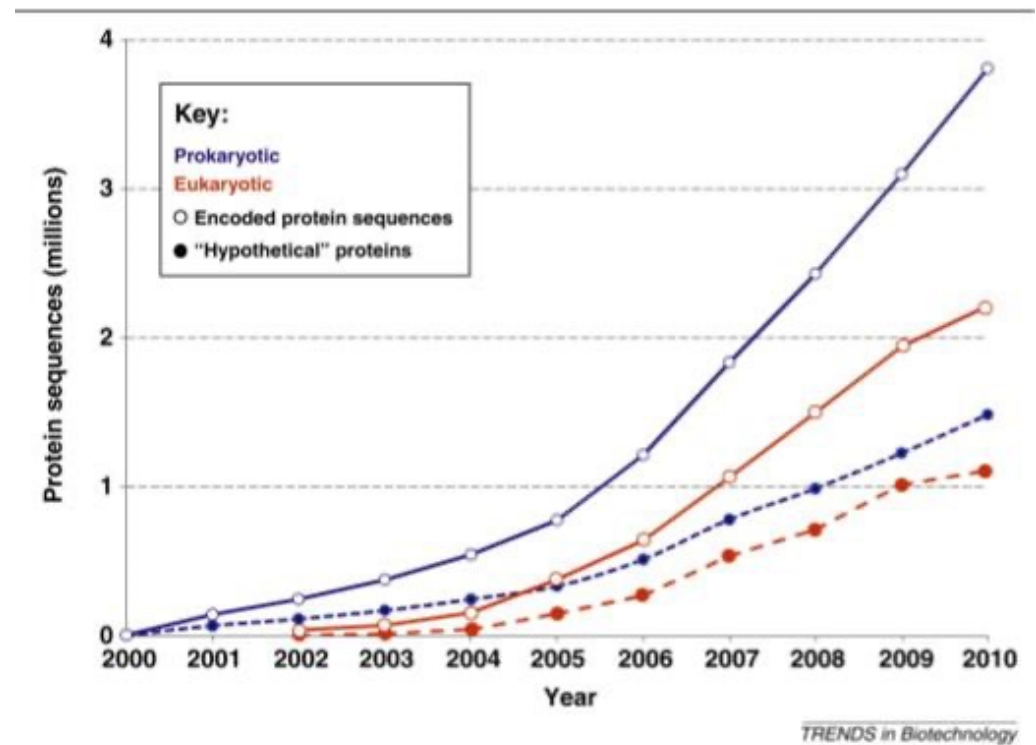
- 4 key functions
 - Storing
 - Querying
 - Integrating
 - Visualizing



Abdulganiyu Abdu Yusuf, Zahraddeen Sufyanu, Kabir Yusuf Mamman & Abubakar Umar Suleiman, "Comparison of Popular Bioinformatics Databases", International Journal of Applied and Advanced Scientific Research, Page Number 19-28, Volume 1, Issue 1, 2016

Data growth in GenBank

Accumulation of protein sequences of unknown function.



NCBI RefSeq database

Real world applications – Medicine

- More drug targets
 - Drugs on the market target only ~500 proteins
- Preventive medicine
 - Development of diagnostic tests to measure a person's susceptibility to different diseases
 - Gene testing
 - pre-symptomatic testing for estimating the risk of developing adult-onset cancers and/or Alzheimer's disease
 - BRCA genetic testing
 - Preventative actions at the earliest possible stages could result in huge advances in our struggle to conquer disease

Real world applications – Medicine

- Gene therapy
 - Gene therapy is the approach used to treat, cure or even prevent disease by changing the expression of a persons genes.

Real world applications – Microbial Genome

- A microscopic organism
- They are found everywhere
 - They are present in the environment, our bodies, the air, food and water.
- Baking, brewing and food industries
- Microbial Genome Project
 - Sequence genomes of bacteria useful in energy production, environmental cleanup, industrial processing and toxic waste reduction

Real world applications – Agriculture

- Sequencing genomes of plants
- Bioinformatic tools can be used to search for the genes within these genomes and to elucidate their functions.
- This specific genetic knowledge could then be used to produce stronger, more drought, disease and insect resistant crops.



Real world applications – Animals

- Sequencing projects of many farm animals – cows, pigs and sheep.
- Understanding of the biology of these organisms will have huge impacts for **improving the production and health** of livestock and ultimately have **benefits for human nutrition**



Genetically Modified Cows Will Produce Low Lactose Milk?

Computers and computer science

- Bioinformatics would not be possible without advances in computing hardware and software.
- Consider the following biological problem:
“Retrieve from a database all sequences similar to a probe sequence”
- A good solution of this problem would appeal to computer science for:

Computers and computer science (cont' d)

- *Analysis of algorithms.* An algorithm is a sequence of instructions that one must perform in order to solve a well-formulated problem.
 - For the retrieval of similar sequences, we need to measure the similarity of the probe sequence to every sequence in the database.
 - It is possible to do much better than the naïve approach of checking every pair of positions in every possible juxtaposition, a method that even without allowing gaps would require a time proportional to the product of the number of characters in the probe sequence times the number of characters in the database.

Computers and computer science (cont' d)

- *Data structures and information retrieval.* How can we organize our data for efficient response to queries?
 - For instance, are there ways to index or otherwise preprocess the data to make our sequence-similarity searches more efficient?
 - How can we provide interfaces that will assist the user in framing and executing queries?

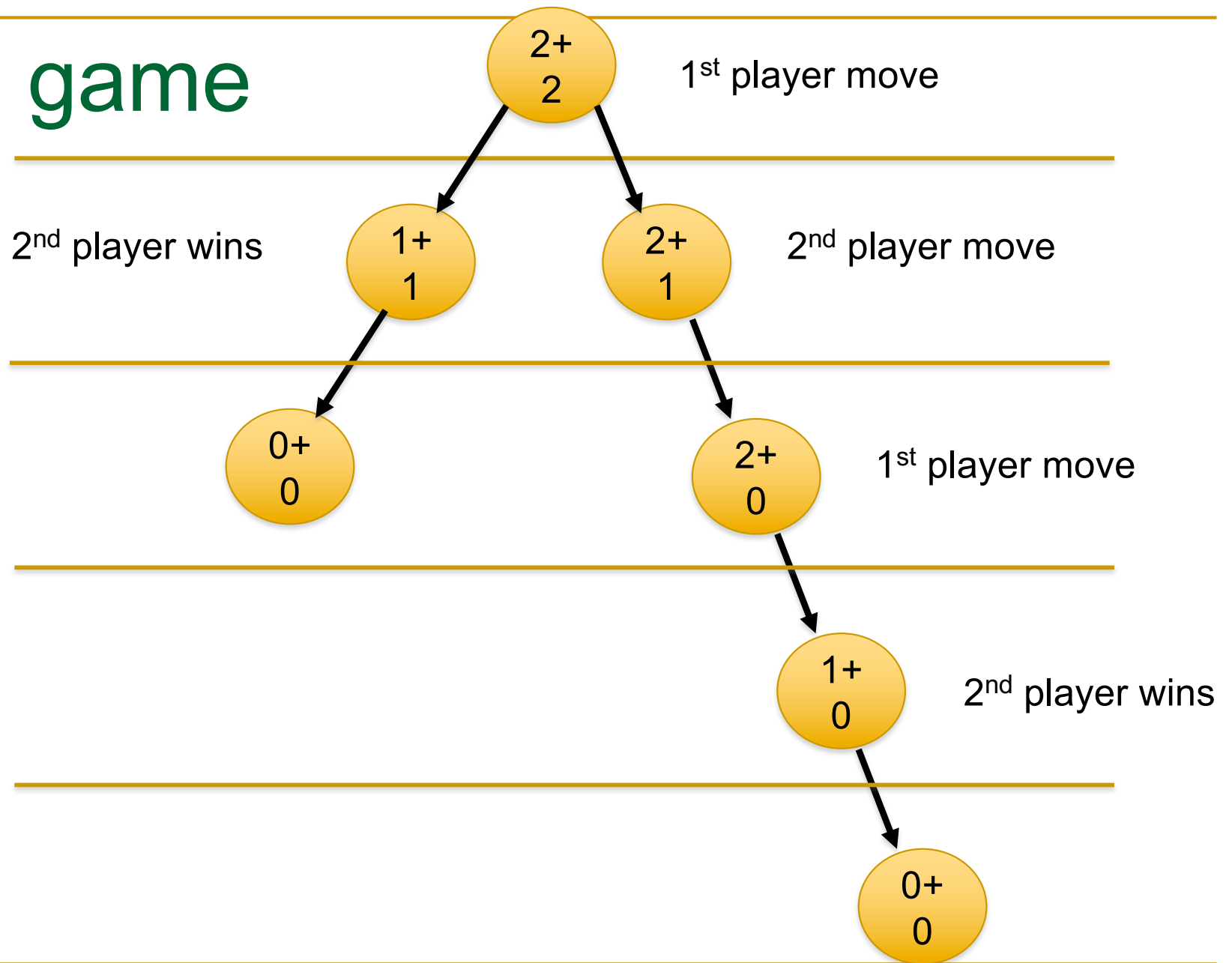
Computers and computer science (cont' d)

- *Software engineering.* Programmers work in high-level languages, such as C, C++, PERL, Python, Java.
- The choice of programming language depends on the nature of the algorithm and associated data structure, and the expected use of the program.

10+10 game

- Two players and two piles of ten rocks.
- Each turn a player may either take one rock from a single pile, or one rock from both piles.
- Once the rocks are taken, they are removed from play.
- The player that takes the last rock wins the game.

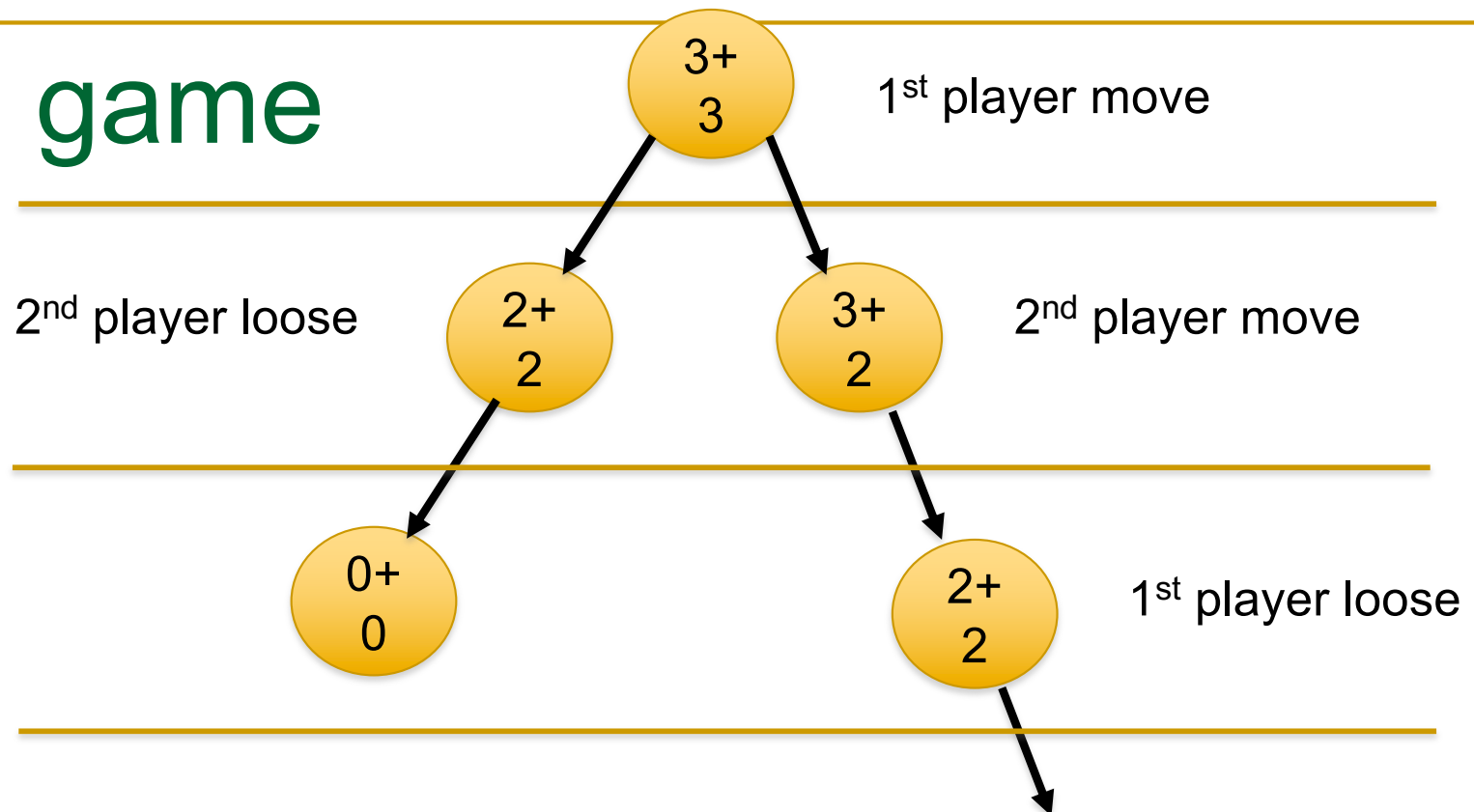
2+2 game



2+2 game

- In any 2+2 game the second player always win.
 - “winning recipe”
 - If first player takes one rock from each pile, the second player takes the remaining rocks and win.
 - If first player takes one rock, the second player will take one rock from the same pile. Then, the first player’s only choice will be to take one rock, and the second player will take the remaining rock and win.

3+3 game



“winning recipe” for the 10+10 game

- (i,j)-entry describes the moves that a player will take in the $i+j$ game, with i and j rocks in piles A and B respectively, for a winning strategy.

←, takes a rock from pile B

↑, takes a rock from pile A

↖, takes one rock from each pile

*, player will lose the game

	0	1	2	3	4	5	6	7	8	9	10
0	*	←	*	←	*	←	*	←	*	←	*
1	↑	↖	↑	↖	↑	↖	↑	↖	↑	↖	↑
2	*	←	*	←	*	←	*	←	*	←	*
3	↑	↖	↑	↖	↑	↖	↑	↖	↑	↖	↑
4	*	←	*	←	*	←	*	←	*	←	*
5	↑	↖	↑	↖	↑	↖	↑	↖	↑	↖	↑
6	*	←	*	←	*	←	*	←	*	←	*
7	↑	↖	↑	↖	↑	↖	↑	↖	↑	↖	↑
8	*	←	*	←	*	←	*	←	*	←	*
9	↑	↖	↑	↖	↑	↖	↑	↖	↑	↖	↑
10	*	←	*	←	*	←	*	←	*	←	*

Why do we care about a rocks game?

- The rocks game is in fact the *sequence alignment* problem.
- The computational idea used to solve both problems is the same.